I. INTRODUCTION

Road transport is an important factor in the economic and social growth of Sri Lanka. However, the traffic law enforcement and driving license management of the country still rely heavily on manual procedures, such as license verification in physical form, paper documentation, and traditional fine management procedures. Such practices can lead to administrative inefficiencies, delays in service delivery, challenges in verifying identities and an increased risk of fraud and data discrepancies. The number of registered vehicles and road users increases, there is a greater need for a more secure, efficient and technology-driven approach to traffic management.

Recent advancements in Artificial Intelligence (AI), biometric authentication, and digital governance have created opportunities to modernize transportation systems worldwide. AI-powered solutions such as facial recognition, driver behavior analysis, and intelligent violation management have demonstrated significant potential in improving road safety, enhancing enforcement efficiency, and reducing human intervention. Many countries have successfully adopted digital licensing platforms that enable secure identity verification, real-time access to driver records, and streamlined administrative processes.

Sri Lanka is still in the midst of its digital transformation initiatives and the comprehensive smart driving license system that incorporates AI technologies is yet to be available. Existing systems are fragmented which hinder effective communication between licensing authorities, traffic police and drivers. Furthermore, the absence of smart monitoring and centralized digital services lowers the efficiency of traffic law enforcement and driver accountability.

This research proposes an AI-Based Smart Driving License System for Sri Lanka to address these challenges. The proposed solution leverages facial recognition technology for secure driver verification, behavior analysis to monitor driving-related behaviors, a point-based system for managing violations and fines to ensure accountability, and a combined mobile and web platform for convenient access to licensing services. It is designed to increase transparency, efficiency and accessibility and to contribute to the modernization of traffic management processes.

The main objectives of this research are to develop a secure digital driving license platform, enhance identity verification through AI-based facial recognition, implement an intelligent behavior and violation management framework, and provide a user-friendly mobile and web environment for both drivers and enforcement authorities. Through these contributions, the proposed system seeks to support Sri Lanka’s transition toward a smarter, safer, and more digitally connected transportation ecosystem.

The remainder of this paper is organized as follows. Section II presents the literature review of relevant technologies and existing studies. Section III discusses the research methodology and system architecture. Section IV describes the implementation of the proposed system. Section V presents the results and evaluation, while Section VI concludes the paper and outlines future research directions.

II. LITERATURE REVIEW

A. Overview

Artificial Intelligence (AI) has become increasingly important in transportation and public service systems due to its ability to improve efficiency, transparency, and operational effectiveness. In Sri Lanka, traffic law enforcement and driving license management continue to rely largely on manual processes, resulting in administrative delays, inconsistencies, and limited accountability [1], [2]. Promising progress has been made in overcoming these challenges through smart transportation solutions utilizing digital technologies, biometric authentication and automated enforcement mechanisms [7]. The literature review focuses on the core technologies and concepts that can be applied to the proposed Smart Driving License System. The concepts include digital licensing, facial recognition, behavior analysis, violation management based on points, and mobile-web platforms.

B. Smart Driving License Systems

Smart driving license systems integrate digital identity management, violation tracking, and administrative services into a unified platform. Research indicates that such systems improve record management, reduce fraudulent activities, and enhance the efficiency of traffic enforcement by maintaining centralized driver databases [15], [37]. Furthermore, digital licensing platforms support online services such as license issuance, renewal, and verification, reducing dependence on paper-based processes [16]. Studies emphasize that successful implementation requires interoperability between government agencies, secure data management practices, and strong governance frameworks, particularly in developing countries undergoing digital transformation [1], [18].

C. Facial Recognition for Identity Verification

Facial recognition technology is one of the most powerful biometrics for identity verification. Approaches based on deep learning such as convolutional neural networks (CNNs) and FaceNet architectures have been used to achieve high recognition accuracy in a variety of applications such as law enforcement, digital identity systems and mobile authentication [5], [11]. Facial recognition allows secure driver authentication in smart licensing environments by matching a captured facial image to registered records, thereby reducing impersonation and identity fraud [3], [10]. Mobile-based facial recognition solutions have improved accessibility and real-time verification [13], [14]. However, previous research points out issues with lighting conditions, facial occlusion, algorithmic bias, and privacy protection that need to be carefully taken into account for system construction [4], [17].

D. Driver Behavior Analysis

Behavior analysis has become an important component of intelligent transportation systems due to its ability to identify unsafe driving patterns and support proactive road safety initiatives. Machine learning and predictive analytics techniques have been successfully applied to analyze driver behavior, violation trends, and risk factors associated with traffic incidents [6], [8]. Research demonstrates that behavioral data can be used to identify repeat offenders and support evidence-based decision-making within enforcement agencies [9]. Furthermore, advanced driver assistance and monitoring systems have shown positive impacts on driving performance and safety awareness [12], [19], [20]. These findings suggest that behavior analysis can contribute significantly to improving accountability and reducing traffic violations.

E. Point-Based Violation and Fine Management Systems

Point-based violation systems assign demerit points to drivers based on traffic offenses and impose escalating penalties when predefined thresholds are exceeded. Studies indicate that such systems promote responsible driving behavior by increasing awareness of the consequences associated with repeated violations [9]. When integrated with digital platforms, point-based systems enable automatic recording of offenses, real-time updating of driver histories, and transparent fine management procedures [18], [21]. Digital fine management solutions also simplify payment processing, improve record accuracy, and reduce administrative burdens on traffic authorities. As a result, these systems are increasingly recognized as effective tools for strengthening road safety and regulatory compliance [6].

F. Mobile and Web-Based Platforms

Mobile and web applications serve as the primary interfaces through which drivers and enforcement officers interact with modern licensing systems. Existing studies show that these platforms support a range of services including digital license display, online renewals, violation notifications, fine payments, and driver record management [16], [21]. For enforcement personnel, mobile applications facilitate rapid identity verification and immediate access to driver information in field environments [13]. Research further emphasizes the importance of responsive design, secure authentication mechanisms, and accessibility features to ensure widespread adoption, particularly in developing-country contexts where connectivity and device limitations may exist [1], [7], [18].

G. Research Gap

Although previous studies have investigated digital licensing systems, facial recognition technologies, behavior analysis, and point-based enforcement mechanisms individually, limited research has focused on integrating these components into a single comprehensive platform. Existing solutions often operate as separate systems, reducing efficiency and limiting interoperability between licensing authorities and enforcement agencies [1], [21]. Furthermore, there is a lack of research specifically addressing the implementation of such integrated solutions within the Sri Lankan context, where traffic management remains predominantly manual [2]. This research addresses these gaps by proposing a Smart Driving License System that combines facial recognition, behavior analysis, point-based violation management, and mobile-web accessibility within a unified platform tailored to local requirements.

H. Summary

The literature reviewed indicates that facial recognition, behavior analysis, digital licensing and point-based enforcement systems have the potential to significantly improve traffic management and road safety. Mobile and web technologies support accessibility and operational efficiencies by providing real-time delivery of services and enforcement support. However, the absence of integrated solutions and limited research on Sri Lanka shows the need for a comprehensive Smart Driving License System. To this end, the present research aims to develop a secure, intelligent and user-centered platform for the modern traffic law enforcement and license management.

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